Oracle Utilities Application SGG v25.10HOT

Product Fix Design

Bug 38950205

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Oracle Utilities Service and Measurement Data Foundation Product Fix Design

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Table of Contents

Table of Contents 3

Reported Issue 4
- Description 4
- Steps To Recreate 4

Investigation 4
- Problem Cause 4
- Suggested Solution 4

Solution Design 4
- Summary 5
- BS/Zone 5
  - D1-ACTHIERQ - Activity Hierarchy Query (Existing) 5
- Scripts 7
  - D1-ACTHIER – Activity Hierarchy (Existing) 7
  - D1-BLDACTHIE – Build Activity Hierarchy (Existing) 8

Recommended Test Cases 9
- Test Case 1: Activity Hierarchy – Field Activity– No Request Orchestrator 9
- Test Case 2: Activity Hierarchy – Field Activity + Existing Request Orchestrator 9
- Test Case 3: Activity Hierarchy – Update Orchestration 10
- Test Case 4: Activity Hierarchy – Cancel Orchestration 10

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1. Reported Issue

Description

When updating standalone Field Activities (no parent) the Update Orchestrator created is not linked to Field Activity. Update orchestrator is created but not shown when viewing the originating activity.

Steps To Recreate

1. Initiate FA/D1-ReadMeter for current date and SP ID 12345 (address = LL11)
2. FA initiated, FA O/B communication is created and moved to Awaiting Response.
3. On FA page click on Initiate Update.
4. Fill out required fields and save.
5. Update Orchestration activity is created.
6. Navigate back to Field Activity.
7. Issue: no Update Orchestration activity listed in Hierarchy tree.
8. The user must search for Update Orchestration activity.

Investigation

Problem Cause

The current system has a limitation where it only populates Orchestration activities related to "Request Orchestrator activities". The defect requires enhancing the system components to also populate in the activity tree, cancel, or update Orchestration activities related to "Field Activity".

Suggested Solution

The fix is to provide functionality to populate. Cancel\Update Orchestration activities related to Field Activity.

Solution Design

.

Summary

Change Type	Y/N
Does this fix require any Admin (Soft) Table Changes to be made by the client?	N
Does this fix require upgrade data changes to be applied?	N
Does the fix involve DB Changes?	N
Is there an impact on any user interface or any other on-line behavior?	Y
Is there an impact on any batch process?	N
Is there an impact on CM – Custom Modification processing?	Y
Does this fix require an application viewer update for Algorithm data (run F1-AVALG)?	N
Does this fix require an application viewer update for MO data (run F1-AVMO)?	N
Does this fix require an application viewer update for Table data (run F1-AVTBL)?	N

BS/Zone

D1-ACTHIERQ - Activity Hierarchy Query (Existing)

```
with
OA as /* Request Orchestrators */
                (select 'OA' LVL, 'D1-ACT' FKREF, '' CAT, '' PARENT,  OA.D1_ACTIVITY_ID ID, oa.bus_obj_cd,
oa.bo_status_cd, oa.cre_dttm
                from D1_ACTIVITY OA, D1_ACTIVITY_TYPE AT
                where OA.D1_ACTIVITY_ID = :F1
                and AT.ACTIVITY_TYPE_CD = OA.ACTIVITY_TYPE_CD
                and AT.ACTIVITY_TYPE_CAT_FLG in ('D1RO','D1MD')),
UO as /* Update Orchestrators */
                 (select 'UO' LVL, 'D1-ACT' FKREF, '' CAT, RA.REL_ACTIVITY_ID PARENT,
SA.D1_ACTIVITY_ID ID, SA.bus_obj_cd, SA.bo_status_cd, SA.cre_dttm
                 from D1_ACTIVITY SA, D1_ACTIVITY_REL RA
                 where RA.REL_ACTIVITY_ID in (select ID from OA)
                 and RA.ACTIVITY_REL_TYPE_FLG = 'D1OA'
                 and SA.D1_ACTIVITY_ID = RA.D1_ACTIVITY_ID),
SA as /* Specific Activities */
                (select 'SA' LVL, 'D1-ACT' FKREF, (select ACTIVITY_TYPE_CAT_FLG
                                        from D1_ACTIVITY_TYPE
                                        where ACTIVITY_TYPE_CD = SA.ACTIVITY_TYPE_CD) CAT,
                                                (select RA.REL_ACTIVITY_ID
                    from D1_ACTIVITY_REL RA
                    where RA.D1_ACTIVITY_ID = SA.D1_ACTIVITY_ID and ACTIVITY_REL_TYPE_FLG =
'D1PR') PARENT,
```

```
                                                                    SA.D1_ACTIVITY_ID ID, SA.bus_obj_cd,
SA.bo_status_cd, SA.cre_dttm
                                                                    from D1_ACTIVITY SA where
SA.D1_ACTIVITY_ID in (select D1_ACTIVITY_ID from D1_ACTIVITY_REL
                        where REL_ACTIVITY_ID in (select ID from OA) and ACTIVITY_REL_TYPE_FLG =
'D1PR'
                        union select :F1 from DUAL where :F1 not in (select ID from OA)) ),
OM as /* Maintenance Orchestration*/
                    (select 'OM' LVL, 'D1-ACT' FKREF, '' CAT, RA.REL_ACTIVITY_ID PARENT,
                     SA.D1_ACTIVITY_ID ID, SA.bus_obj_cd, SA.bo_status_cd, SA.cre_dttm
                     from D1_ACTIVITY SA, D1_ACTIVITY_REL RA
                     where RA.REL_ACTIVITY_ID in (select ID from SA)
                     and RA.ACTIVITY_REL_TYPE_FLG = 'D1SA'
                     and SA.D1_ACTIVITY_ID = RA.D1_ACTIVITY_ID),
OC as /* Outbound Communication */
                    (select 'OC' LVL, 'D1-CMOUT' FKREF, '' CAT, (
                        select PK_VALUE1 from D1_COMM_OUT_REL_OBJ
                        where D1_COMM_ID = OC.D1_COMM_ID
                        and COMM_REL_OBJ_TYPE_FLG = 'D1PA') PARENT,
                    OC.D1_COMM_ID ID, OC.bus_obj_cd, OC.bo_status_cd, OC.cre_dttm
                    from D1_COMM_OUT OC
                    where OC.D1_COMM_ID in (
                    select RO.D1_COMM_ID
                    from D1_COMM_OUT_REL_OBJ RO
                    where RO.PK_VALUE1 in (
                    select ID from SA union select ID
                    from UO union select ID from OM)
                    and RO.COMM_REL_OBJ_TYPE_FLG = 'D1PA'
                    union select :F2 from DUAL) ),
IC as /* Inbound Communication */
                    (select 'IC' LVL, 'D1-CMIN' FKREF, '' CAT, (select PK_VALUE1 from D1_COMM_IN_REL_OBJ
where D1_COMM_ID = IC.D1_COMM_ID and COMM_REL_OBJ_TYPE_FLG = 'D1OB') PARENT,
                    IC.D1_COMM_ID ID, IC.bus_obj_cd, IC.bo_status_cd, IC.cre_dttm   from D1_COMM_IN IC
                    where IC.D1_COMM_ID in (select RO.D1_COMM_ID from D1_COMM_IN_REL_OBJ RO
                    where RO.PK_VALUE1 in (select ID from OC)
                    and RO.COMM_REL_OBJ_TYPE_FLG = 'D1OB'
                    union select :F3 from DUAL) ),
 CE as /* Completion Event under I/B Comm */
                    (select 'CE' LVL, 'D1-CEVT' FKREF, '' CAT, CE.IN_COMM_ID PARENT,
                     CE.COMPL_EVT_ID ID, CE.bus_obj_cd, CE.bo_status_cd, CE.cre_dttm  from
D1_COMPL_EVT CE
                     where CE.IN_COMM_ID in (select ID from IC)
                     and CE.IN_COMM_ID is not null
                     and CE.D1_ACTIVITY_ID in (select ID from SA)
                     and CE.D1_ACTIVITY_ID is not null),
CEA as /* Completion Event under Activity */
                    (select 'CE' LVL, 'D1-CEVT' FKREF, '' CAT, CE.D1_ACTIVITY_ID PARENT,
CE.COMPL_EVT_ID ID, CE.bus_obj_cd, CE.bo_status_cd, CE.cre_dttm
                    from D1_COMPL_EVT CE  where CE.COMPL_EVT_ID not in (select ID from CE)  and
CE.D1_ACTIVITY_ID in (select ID from SA) and CE.D1_ACTIVITY_ID is not null)
                     select LVL, FKREF, CAT, PARENT, ID, Q.bus_obj_cd, Q.bo_status_cd, Q.cre_dttm, (
                     SELECT SO.BO_OPT_VAL FROM F1_BUS_OBJ BO, F1_BUS_OBJ_STATUS_OPT SO
                     WHERE BO.BUS_OBJ_CD = Q.BUS_OBJ_CD
                     AND SO.BUS_OBJ_CD = BO.LIFE_CYCLE_BO_CD
                     AND SO.BO_OPT_FLG = 'D1LS'
                     AND SO.BO_STATUS_CD = Q.BO_STATUS_CD) LOG_ST,
```

```
(SELECT SO.BO_OPT_VAL FROM F1_BUS_OBJ BO, F1_BUS_OBJ_STATUS_OPT SO  WHERE
BO.BUS_OBJ_CD = Q.BUS_OBJ_CD AND SO.BUS_OBJ_CD = BO.LIFE_CYCLE_BO_CD AND
SO.BO_OPT_FLG = 'D1CL' AND SO.BO_STATUS_CD = Q.BO_STATUS_CD) COLOR
from (select * from OA union
     select * from UO union
     select * from SA union
            select * from OM union
     select * from OC union
     select * from IC union
     select * from CE union
     select * from CEA) Q
order by cre_dttm
```

Scripts

D1-ACTHIER – Activity Hierarchy (Existing)

Step 40

```
// assume blank UO2 Comm
if ("count(D1-ACTHIERQ/rows[level='OM']/id) = 0")
  move 'OM' to "D1-ACTHIERQ/+rows/level";
end-if;

for ($r in "D1-ACTHIERQ/rows[level='OM']")
   move "string($r/parentId)" to $p;
   move "$r" to "parm/orchestrator/activity[id=$p]/+activity";
end-for;
```

```
Schema
<schema type="schema">
   <activityId dataType="string"/>
   <communicationId dataType="string"/>
   <maintenanceObject dataType="string"/>
   <orchestrator type="list">
      <id fkRef="D1-ACT" dataType="string"/>
      <bo dataType="string"/>
      <logicalStatus dataType="string"/>
      <fkReference dataType="string"/>
      <info dataType="string"/>
      <color dataType="string"/>
      <activity type="list">
         <id fkRef="D1-ACT" dataType="string"/>
         <bo dataType="string"/>
         <logicalStatus dataType="string"/>
         <fkReference dataType="string"/>
         <info dataType="string"/>
         <color dataType="string"/>
         <activity type="list">
            <id fkRef="D1-ACT" dataType="string"/>
```

```
            <bo dataType="string"/>
            <logicalStatus dataType="string"/>
            <fkReference dataType="string"/>
            <info dataType="string"/>
            <color dataType="string"/>
            <commOut type="list">
               <id fkRef="D1-CMOUT" dataType="string"/>
               <bo dataType="string"/>
               <logicalStatus dataType="string"/>
               <fkReference dataType="string"/>
               <info dataType="string"/>
               <color dataType="string"/>
               <commIn type="list">
                  <id fkRef="D1-CMIN" dataType="string"/>
                  <bo dataType="string"/>
                  <logicalStatus dataType="string"/>
                  <fkReference dataType="string"/>
                  <info dataType="string"/>
                  <color dataType="string"/>
               </commIn>
            </commOut>
         </activity>
         <completionEvent type="list">
            <id fkRef="D1-CEVT" dataType="string"/>
            <bo dataType="string"/>
            <logicalStatus dataType="string"/>
```

D1-BLDACTHIE – Build Activity Hierarchy (Existing)

Step 20

```
for($activity in "$orchestrator/activity")
    if ("string($activity/id) != $BLANK")
     move "$activity/id" to "parm/+children/attr/id";
                    :
                    :
                    :
if("string($orchestrator/fkReference) = $BLANK")
for($om_activity in "$activity/activity")
        move "$om_activity/id" to "parm/+children/attr/id";
        move "$om_activity/info" to "parm/children[last()]/attr/infoString";
        move "$om_activity/fkReference" to "parm/children[last()]/attr/fkReference";
        // Get nav option
        move null to "F1-GetForeignKeyReference";
        move "$om_activity/fkReference"  to "F1-GetForeignKeyReference/input/fkReference";
        move "$om_activity/id" to "F1-GetForeignKeyReference/input/fkValue1";
        invokeBS 'F1-GetForeignKeyReference' using "F1-GetForeignKeyReference";
        move "F1-GetForeignKeyReference/output/navigationOption" to "parm/children[last()]/attr/navOption";
        move "$om_activity/color" to "parm/children[last()]/attr/color";
        move "xs:integer($nodeId) +1" to $nodeId;
        move "$nodeId" to "parm/children[last()]/attr/nodeId";
        move "xs:integer($nodeId)" to $om_activityNodeId;
        move "$activityNodeId" to "parm/children[last()]/attr/parentNodeId";
```

```
        move 'true' to "parm/children[last()]/attr/isActivity";
          for($om_commOut in "$om_activity/commOut")
              move "$om_commOut/id" to "parm/+children/attr/id";
              move "$om_commOut/info" to "parm/children[last()]/attr/infoString";
              move "$om_commOut/fkReference" to "parm/children[last()]/attr/fkReference";
              // Get nav option
              move null to "F1-GetForeignKeyReference";
              move "$om_commOut/fkReference"  to "F1-GetForeignKeyReference/input/fkReference";
              move "$om_commOut/id" to "F1-GetForeignKeyReference/input/fkValue1";
              invokeBS 'F1-GetForeignKeyReference' using "F1-GetForeignKeyReference";
              move "F1-GetForeignKeyReference/output/navigationOption" to
"parm/children[last()]/attr/navOption";
              move "$om_commOut/color" to "parm/children[last()]/attr/color";
              move "xs:integer($nodeId) +1" to $nodeId;
              move "$nodeId" to "parm/children[last()]/attr/nodeId";
              move "xs:integer($nodeId)" to $om_commOutNodeId;
              move "$om_activityNodeId" to "parm/children[last()]/attr/parentNodeId";
              move 'true' to "parm/children[last()]/attr/isCommOut";
         end-for;
 end-for;
 end-if; //Orchestration Maintanence
```

Recommended Test Cases

Test Case 1: Activity Hierarchy – Field Activity– No Request Orchestrator

1. Repeat Steps To Recreate
2. Observe no issue.

Test Case 2: Activity Hierarchy – Field Activity + Existing Request Orchestrator

1. Repeat Steps To Recreate
2. Observe no issue.

Test Case 3: Activity Hierarchy – Update Orchestration

1. Repeat Steps To Recreate
2. Link into Update Orchestration
3. Observe Update Orchestration highlighted. Full tree populated.

Test Case 4: Activity Hierarchy – Cancel Orchestration

1. Repeat Steps To Recreate
2. Observe Cancel Orchestration populated in activity tree.